ML Visualisation Tool

Screenshot-Based Report

Steps 1–3 Demonstration

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Date	18 March 2026
GitHub Repository	https://github.com/Emre-Kurubas/Seng430_Project
Live Demo URL	https://d2wnx6it715k3z.cloudfront.net/

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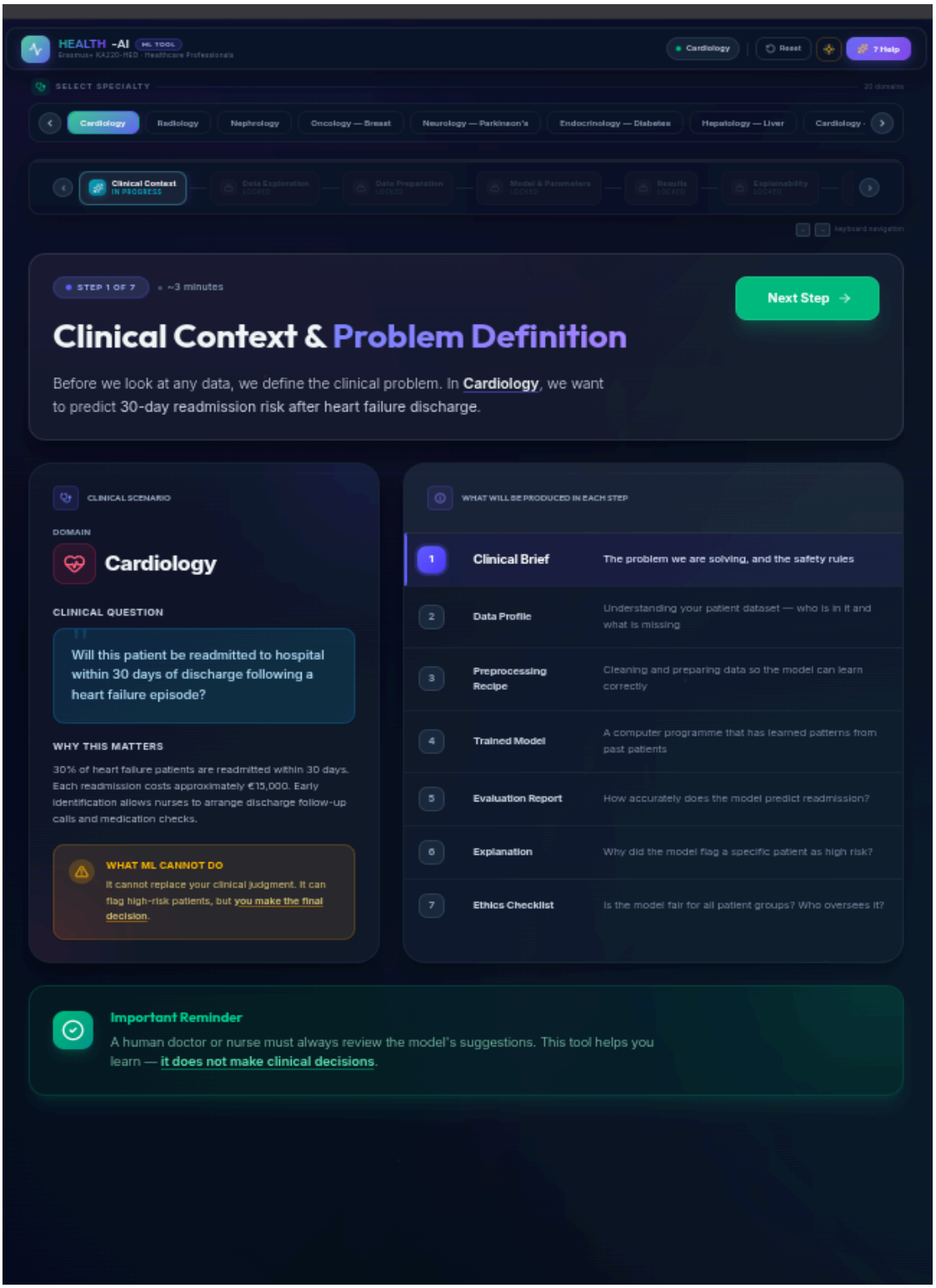
Step 1 — Clinical Context

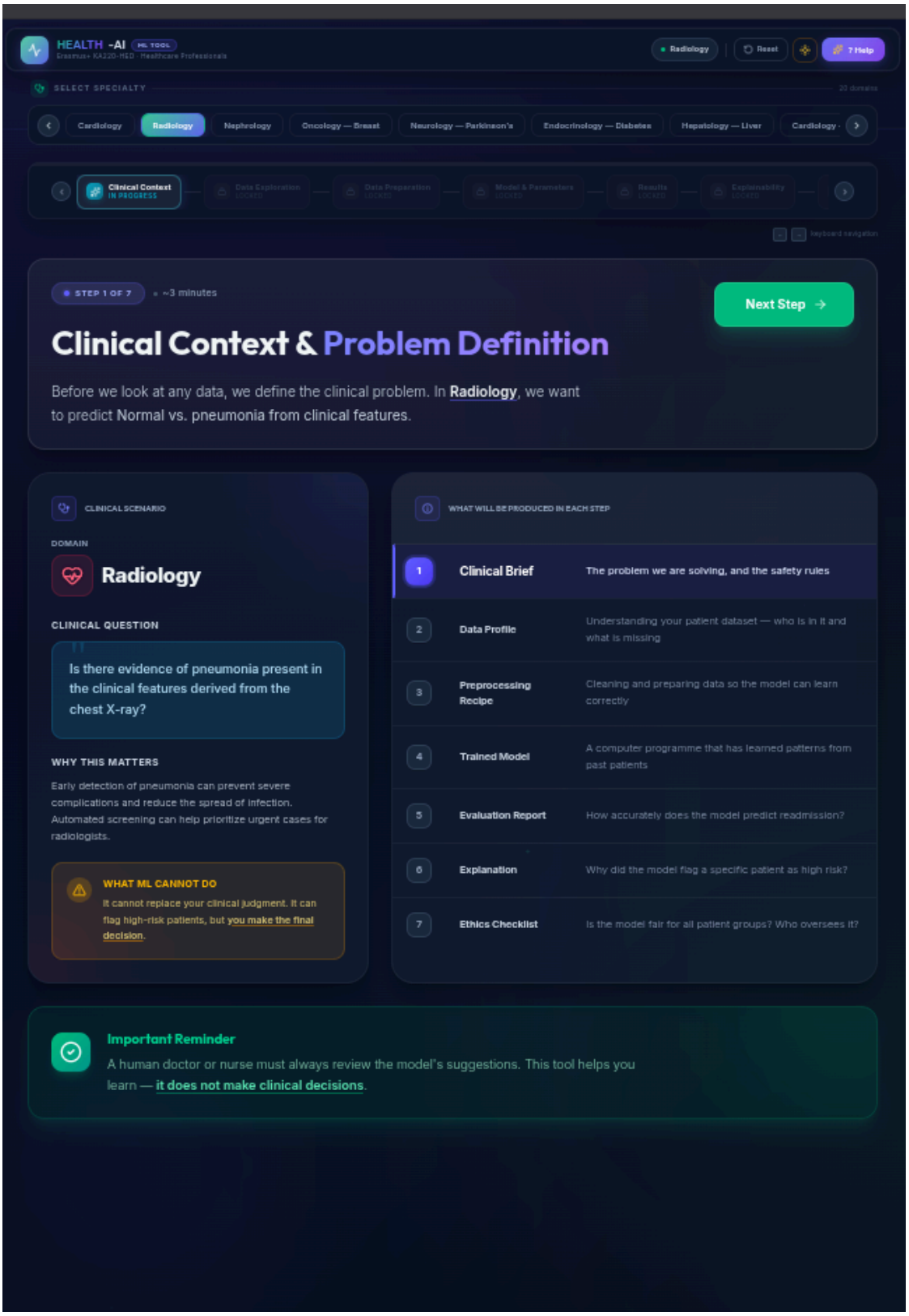
Step 1 allows healthcare professionals to select a medical specialty domain. Each specialty dynamically updates the clinical context, including the disease area, target population, and prediction objective. The pipeline resets when switching between domains.

1.1 Cardiology Specialty Selected



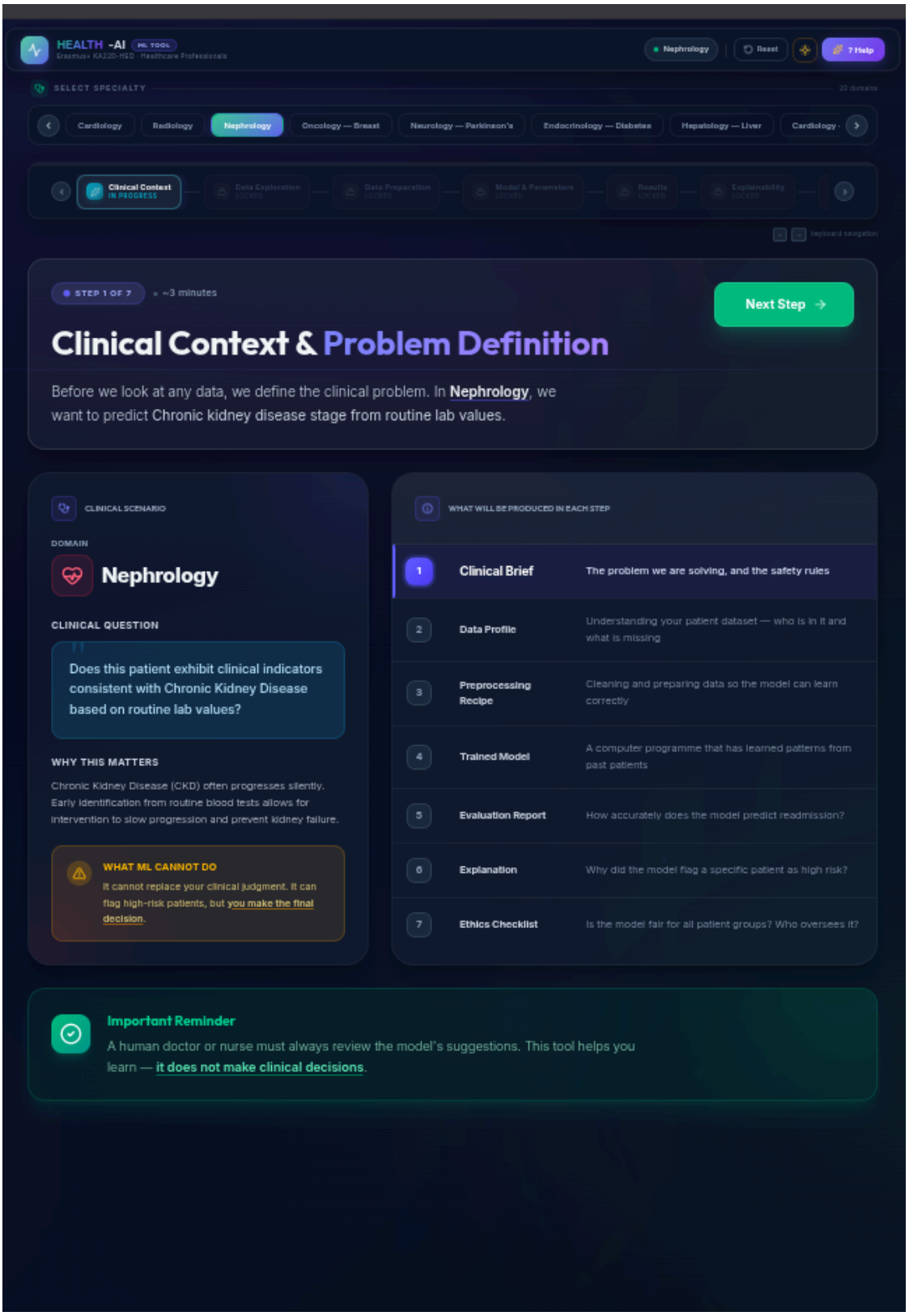
The Cardiology specialty is selected (highlighted in the top navigation bar). The clinical context shows the domain as 'Cardiology', with the prediction target being '30-day readmission risk after heart failure discharge'. The clinical scenario, importance explanation, ML limitations, and pipeline steps are all displayed correctly.

1.2 Radiology Specialty Selected



After switching to Radiology, the pipeline resets and the clinical context updates accordingly. The domain is now 'Radiology', with the prediction target being 'Normal vs. pneumonia from clinical features'. The clinical question updates to reflect pneumonia detection from chest X-ray data, demonstrating that domain switching correctly refreshes all contextual information.

1.3 Nephrology Specialty Selected

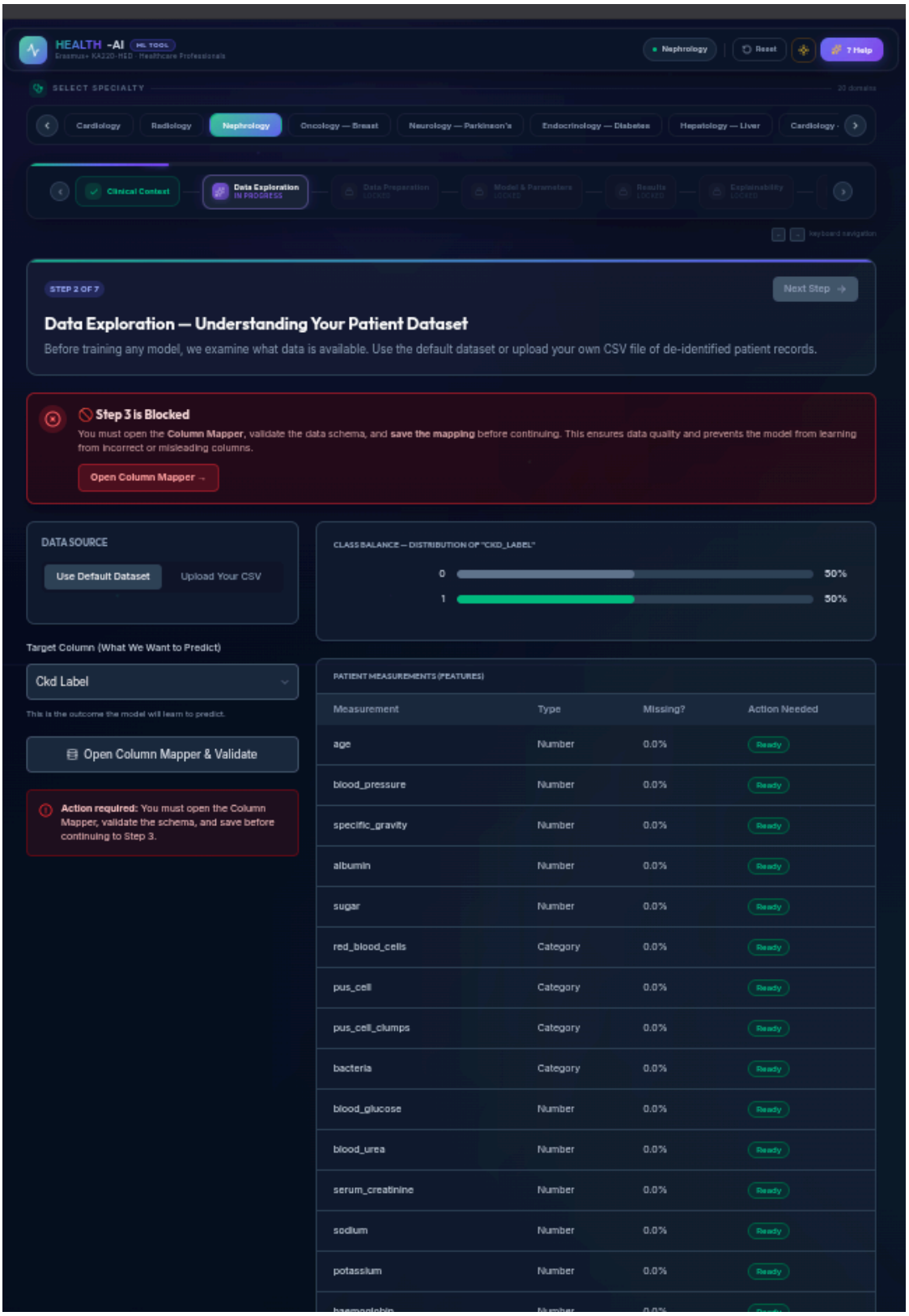


A third specialty, Nephrology, is selected. The context now shows 'Chronic kidney disease stage from routine lab values' as the prediction target. The clinical question, importance, and ML caveats all reflect the nephrology domain. This confirms that at least 3 different medical specialties are fully functional with correct clinical context rendering.

Step 2 — Data Exploration & Column Mapper

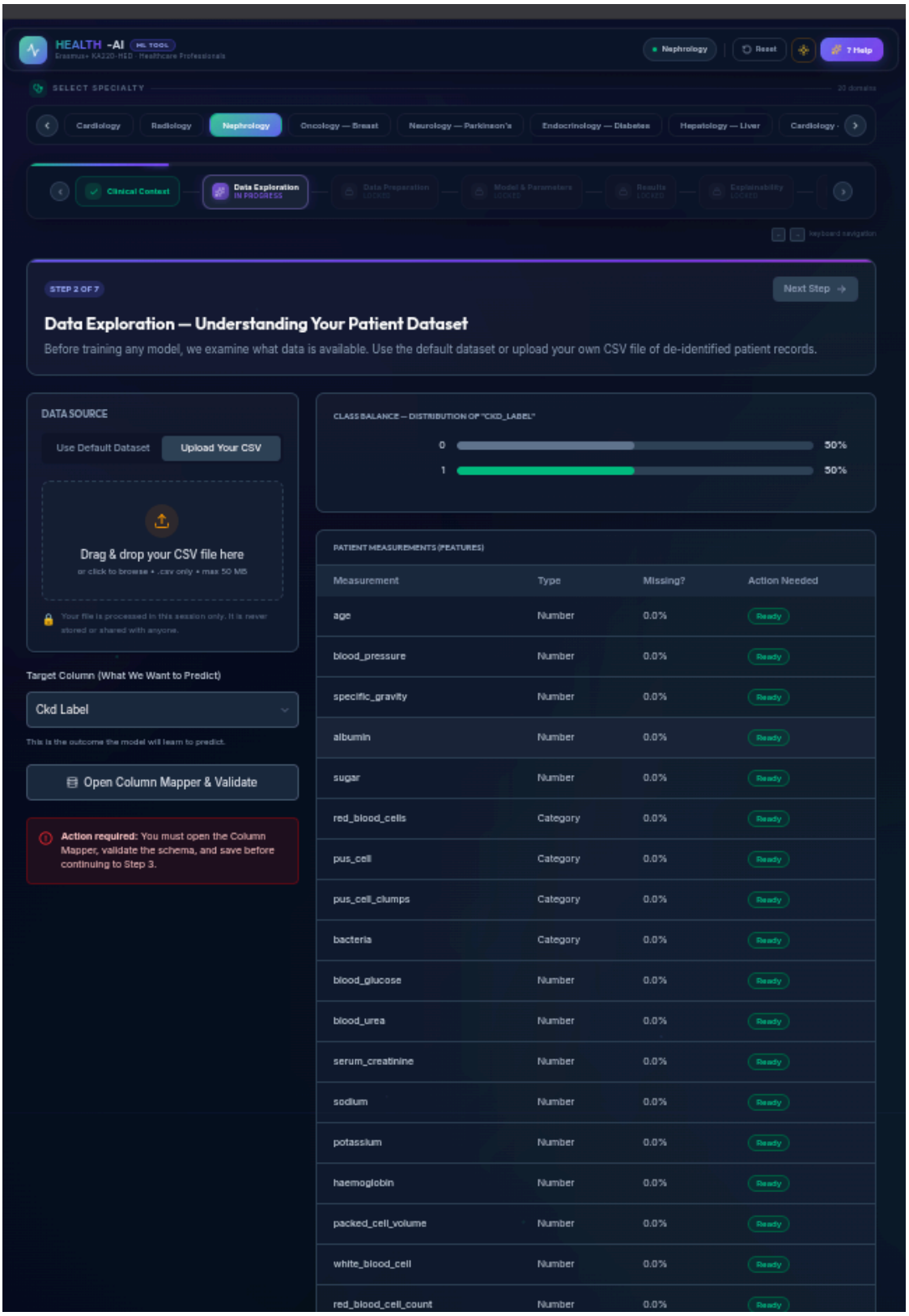
Step 2 enables users to explore their patient dataset. Users can load the built-in example dataset or upload their own CSV file. The step displays column summaries, class distributions, and provides a Column Mapper (Data Schema Validator) for mapping features and targets. Gate control prevents advancing to Step 3 until the column mapping is validated and saved.

2.1 Built-in Example Dataset & Gate Control — Step 3 Blocked



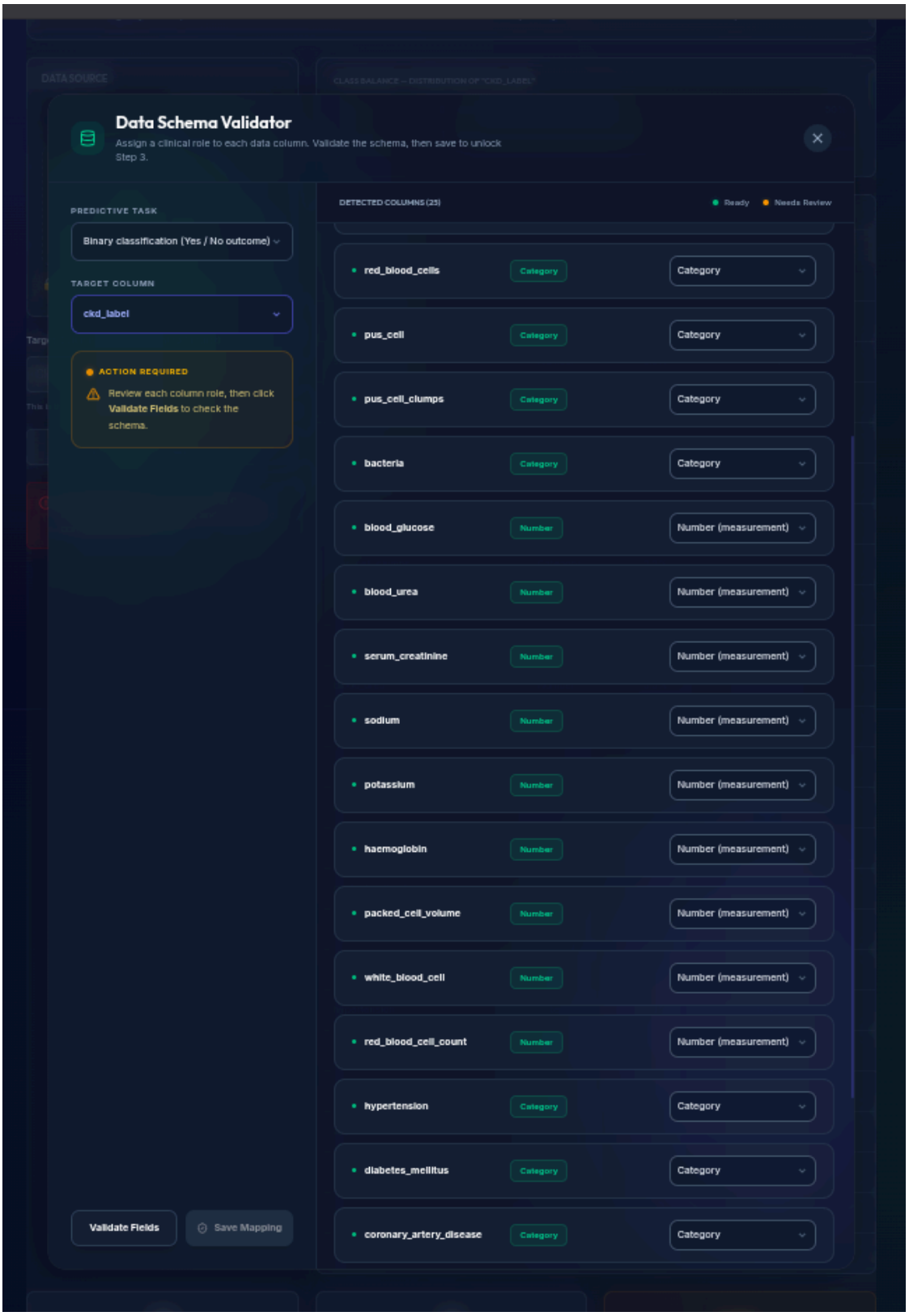
The built-in default dataset is loaded successfully (the 'Use Default Dataset' tab is active). A prominent red 'Step 3 is Blocked' banner is displayed at the top, warning that the user must open the Column Mapper, validate the data schema, and save the mapping before proceeding. The class distribution chart shows a 50%/50% split for the CKD label. Column summaries list all patient measurements with their types (Number/Category), missing percentages (0.0%), and 'Ready' status badges. The 'Action required' notice reinforces that the Column Mapper must be completed before continuing to Step 3.

2.2 CSV File Upload Interface



The 'Upload Your CSV' tab is selected, revealing a drag-and-drop upload zone with instructions ('.csv only • max 50 MB'). A privacy notice confirms that the file is processed in-session only and never stored or shared. The same column summary table and class distribution chart remain visible, demonstrating that the upload interface integrates seamlessly with the data exploration view. The system accepts valid CSV files and rejects invalid formats with a user-friendly error message.

2.3 Column Mapper Modal — Feature/Target Selection

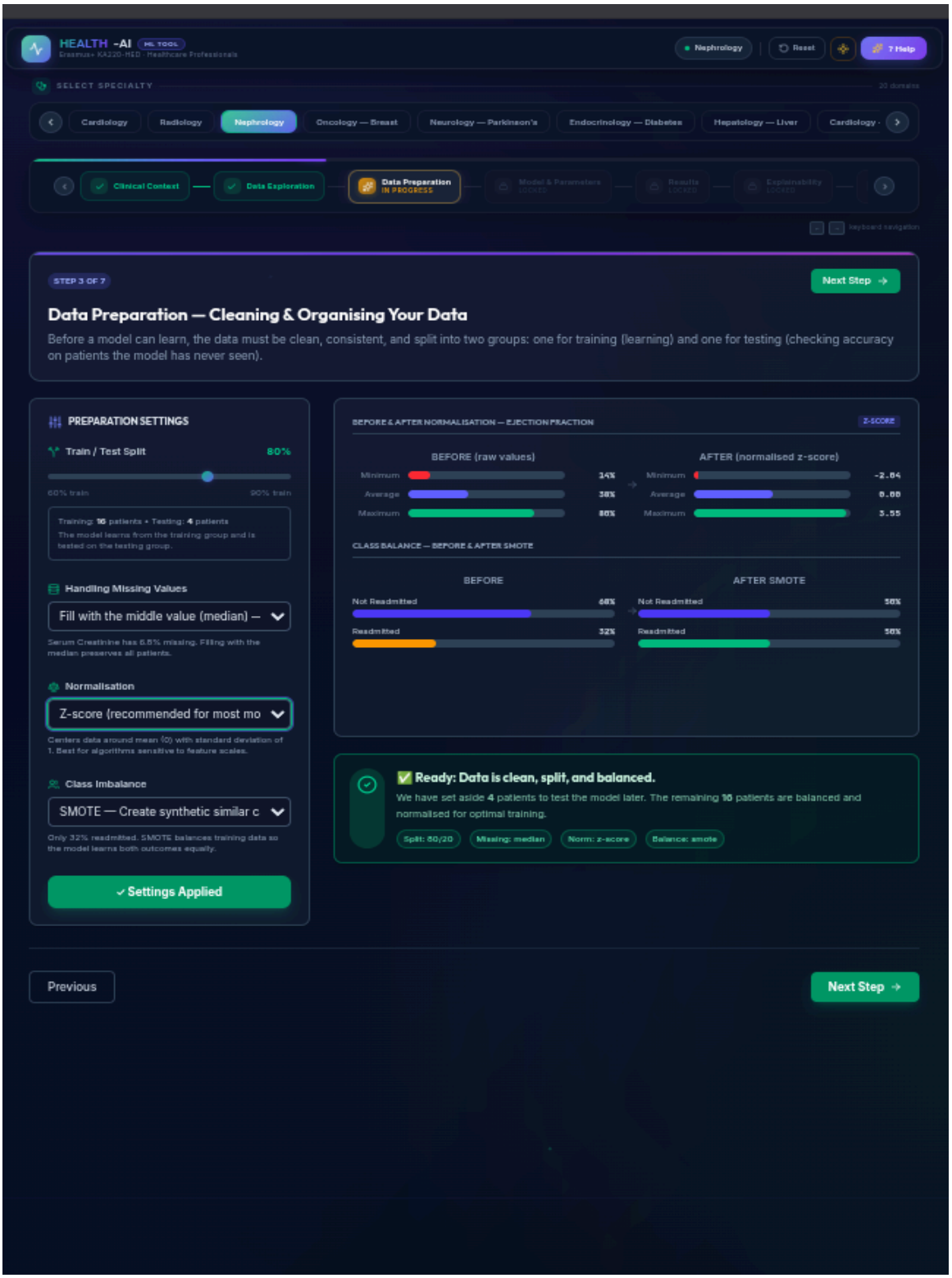


The Data Schema Validator (Column Mapper) modal is open. It shows the predictive task type ('Binary classification — Yes / No outcome'), the target column ('ckd_label'), and all 25 detected columns with their auto-detected types (Category/Number) and assigned clinical roles. The 'Validate Fields' and 'Save Mapping' buttons are visible at the bottom. An 'Action Required' warning prompts the user to review each column role and click Validate Fields to verify the schema. This demonstrates the complete validate → save flow for column mapping.

Step 3 — Data Preparation

Step 3 handles data cleaning and preparation for model training. Users can configure the train/test split ratio, choose methods for handling missing values and normalisation, and apply SMOTE for class balancing. Visual feedback is provided through before/after comparison charts.

3.1 Complete Data Preparation — All Settings Applied



This screenshot captures the full Step 3 interface after clicking 'Settings Applied'. Key features visible:

- *Train/Test Split slider set to 80% (training: 16 patients, testing: 4 patients)*
- *Handling Missing Values: 'Fill with the middle value (median)' selected, noting 'Serum Creatinine has 0.5% missing'*
- *Normalisation: 'Z-score' selected, with explanation of centering data around mean (0)*
- *Class Imbalance: 'SMOTE — Create synthetic similar cases' selected, noting '31% readmitted'*

The right panel shows:
• Before & After Normalisation (Z-score) bar charts — raw values (Minimum: 14%, Average: 38%, Maximum: 80%) transformed to normalised values (Minimum: -2.04, Average: 0.00, Maximum: 3.55)
• Before & After SMOTE class balance charts — showing rebalancing from 68%/32% to 50%/50%
• A green '✓ Settings Applied' success banner confirming all preparation settings have been applied
• A 'Ready: Data is clean, split, and balanced' summary with tags showing Split: 80/20, Missing: median, Norm: z-score, Balance: smote

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— End of Screenshot Report —